Chi-Square Tests

What is Chi_Square Test?
A Chi-Square tests checks whteher the difference betweeen observed frequencies and expected frequencies is statistically significant.

Techincally:
The Chi-Square test is a non-prametric statistical test used to analyze categorical frequency data.
Example:
1. Gender: Male,Female
2. Product preference: A,Bor C
3. Marketing Channel: Online, In-store
4. Customer Status: Purhcased, Not Purhcased

$$x^2 = \sum(O-E)^2/E$$

O --> Observed Frequency (actual count)
E--> Expected Frequency(count expected under H0)
$\sum$ = Add contributions from all categories or cells.

Chi-Square Godness of Fit Test
The Goodness of Fit checks whther observed frequenies follow a specified expected ditribution

Business Usecase: A company wants to know whether customers prefer four products equally.

Step 1: Understand the Data
A survey of 200 customer podces he fllowng results

Product	Obsered(O)	Expected (E)
A	50	50
B	70	50
C	45	50
D	35	50
Total	200	200

If preferenec are equal, each producshould receive:

E = 200/4 =50

Step 2: State the hypothesis

H0 : All four products are equally preferred
H1 : Produc preference are not equally distributed.

Use significane level alpha = 0.05.

Step 3: Calculate Chi-Square: $x^2 = \sum(O-E)^2/E$

Product	Calulation	Contribution
A	$(50-50)^2/50$	0
B	$(70-50)^2/50$	8
C	$(45-50)^2/50$	0.5
D	$(35-50)^2/50$	4.5

Total 13

Therefore

$x^2 = 13.00$

Step 4: Calclulate degrees of freedom

For goodness of Fit

df = k-1

Here, k = 4 == categories.

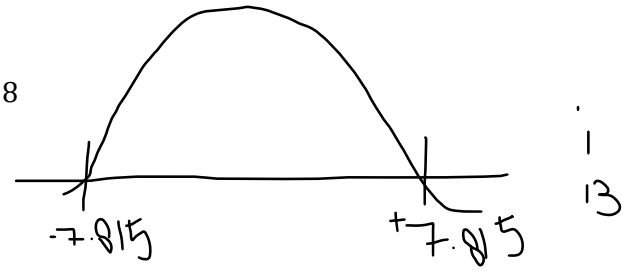
df = 4-1 = 3

Step 5: Find the crirtical Value

alpha = 0.05 and df = 3

7.814728

x^2critical = 7.815



Step 6: Make the Decision

Calculated Statistic	13
Crtitical Value	7.815

13 > 7.815

Reject H0

There is statistically significant evdence that customer preference do not follow equal distributinon

Business Interpretation: Product B receives more selections that expected, while Product B receives fewer.

The company should investigate product demand before making inventory and marketing decisions,

<u>Chi-Square Test of Independence</u>

The Chi-Square Test of Indepnednce determines whether two categorical variables are startistically associated.

Businesss Question: Is product prefernce related to a gender?

Step 1: Observed frequency table

A company surveys 200 customers

Gender	Product A	Product B	Product C	Total
Male	30	50	20	100
Female	20	30	50	100
Total	**50**	**80**	**70**	**200**

Step 2: State the Hypothesis

H0 : Gender and product preference are independent.

H1 : Gender and product preference are associated.

Step 3: Calcuate the Expected frequencies

For Every Cell:

$$E = \frac{\text{Row Total x Column Total}}{\text{Grand Total}}$$

Example: Expected frequency for Male-Product A:

	E = (100*50)/200	25
product B	E = (100*80)/200	40
Product C	E = (100*70)/200	35

Gender	Product A	Product B	Product C
Male	25	40	35
Female	25	40	35

These are the counts if we expect gender and product preference were indpendent.

Step 4 : Calculate each cell's contribution

$(O-E)^2/E$

Male:	
A	1
B	2.5
C	6.4285714
Female	
A	1
B	2.5
C	6.4285714
Total χ^2	19.857143

Step 5: Calculate the Degrees of Freedom

For test independence:

$df = (r-1)(c-1)$

Here , r =2 and c= 3, excluding the totals

$df = (2-1)(3-1) = 2$

$df = 2$

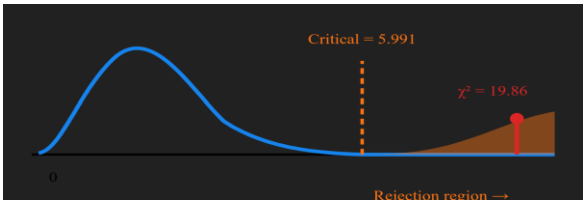
At alpha = 0.05 and df = 2

$\chi 2$ critical = 5.991 5.991465

Decision;

19.857>5.991

Reject H0, There is evidence of association betweeen gender and product preference in this sample.



Expected Frequency and Assumptions

Expected Frequency:

It is the count predicted under the null hypothesis. It is not the actual count and is not a percntage.

For a contingency table:

$$E = \frac{\text{Row Total x Column Total}}{\text{Grand Total}}$$

Assumptions

1. Data consists of counts in categorical groups
2. Observations are independent
3. Sampling is random or appropriately representative.
4. Categories are mutually exclusive and collectively exhastive.
5. Expected frequencies are sufficiently large.

Guideline: Expected frequencies should be atleast 5 in each cell.

Example: A survey of only 12 customers has expected counts of 8, 2 and 2 across three categories. The expectd counts of 2 are too small for usa Chi-Square approximation to be trusted.

Complete Chi-Square Test:

Workflow:

1. Define the business proble
2. State H0 and H1
3. Prepare the observed frequency table
4. Calculate expected frequencies
5. Calcualte $\chi^2 = \sum(O-E)^2/E$
6. Calculate the degree of freedom.
7. Compare the calcualted statistic with the critical value
8. Interpret the result in business terms,

A company claims that customers choose three marketing channels equally. A survey of 150 customers produces these counts.

Channel	Observed	Calcualation $(O-E)^2/E$
Online	70	8
TV	50	0
In-Store	30	8
Total	150	16

E = 150 /3 50

$X^2 = 16$

Degree of freedom = 3-1 = 2

Critical value

X^2 critical = 5.991 5.991465

Since 16 > 5.991, reject H0.

Interpretation:

Customer choices do not follow an equal distribution across the three channels. The marketing Team should investgate why online performs different before allocating its budget.

Excel Funcitons:

Function	Purpose
=CHISQ.TEST(actual_range, expected_range)	Returns the p-value
=CHISQ.INV.RT(alpha,df)	Retutns the right tail critical value
=CHISQ.DIST.RT(x, df)	Returns the right tail probability of a give number

Business Applications

1. Marketing

Are product preferences different across regions?

2. Retail

Is product category associated with store type?

3. HR

Is employee satisfaction associated with a department.

4. Helathcare

IS recovery outcome associated with a treatment type?

5. Finance:

Is loan defualt associated with customer category?

6. Operations:

Do defect types vary across production lines?

Does Chi-Square test tells about causation/ prove causation?

No, It provides evidence of a difference or association, not proof that one variable causes the other.

accident --> treatment at hostpital.

flood--> crops affcted--> food shortage/medical treatment.